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# CACHE TRANSLATION ACROSS MODEL SCALE: AN AUDIT WITH IDENTITY CONTROLS AND TEACHER- CONTENT PROBES

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## ABSTRACT

A KV cache is model-specific, so when several models at different scales serve the same long context, each of them prefills it again. Cross-model cache translation removes the later prefills by mapping the KV cache a large teacher builds over that context into the cache space of a smaller student. Its reported outcomes span near-native retention and above-baseline gain on one side and severe deficits on the other. An end-to-end score conflates three questions that fail separately: whether the injection pipeline consumes a foreign cache correctly (*mechanics*), whether a translator replaces the student’s own cache at replacement level (*mapping*), and whether the teacher’s cache carries content the frozen student can exploit (*exploitability*). We audit all three under strict zero re-prefill on Qwen3-4B→1.7B and three further Qwen3 scale pairs (four in total), using per-sample identity controls, disjoint calibration and evaluation splits, and gold-letter-probability metrics with bootstrap intervals. Mechanics passes to a numerical tolerance: every evaluated sample of the identity injection has logit cosine  $\geq 0.9998$  against the student’s own prefill (observed minimum 0.99986, mean 0.99998), with a change of  $+0.0001$   $[-0.0006, +0.0009]$ . No translator we test replaces the strong student’s own prefill. The audit protocol records 133 translation configurations across all tested pairs; on the primary strong-student pair the largest point estimate is  $-0.138$ , and every interval there lies below zero except the best one, which spans it. The deficit does not come from the calibration budget or from translator capacity: calibration volume from 10 to 500 examples moves the result by 0.02, removing the analytic anchor of our most architecture-aware translator improves perplexity without improving answers, and a 37M-parameter joint MLP does not help. Aligning our implementation with the closest published design—concatenating the selected source layers and calibrating on 1,024-token web-text passages—leaves the verdict unchanged ( $-0.223$  to  $-0.279$ ). The weak 0.6B student reaches  $+0.010$   $[-0.089, +0.106]$ , which establishes neither degradation nor the reporting margin we adopt ( $\epsilon=0.02$ ). Teacher-content probes never produce a gain whose interval excludes zero, and the family-wise 95% bootstrap upper estimate over the six confirmatory probe configurations is  $+0.009$ . Two decouplings survive: better KV reconstruction (held-out  $R^2$  0.854 for keys, 0.481 for values) does not improve answers, and near-native perplexity does not either.

## 1 INTRODUCTION

Serving systems increasingly reuse computation across models: the same long context is prefilled by several models at different scales. Because a KV cache is model-specific, a second model must normally prefill the shared prefix again. Cache translation attacks this by fitting a translator from the teacher’s stored cache into the student’s cache space, after which the student answers a short query from the translated cache alone—*strict zero re-prefill* (Fu et al., 2025; Dery et al., 2026; Lee et al., 2026; Heo et al., 2026; Qu et al., 2026; Li et al., 2026). Reported outcomes now span a wide range: near-native retention on Qwen3 transfer directions (Qu et al., 2026), accuracy above the receiving model’s own baseline on LongBench2 (Li et al., 2026), and large deficits elsewhere.

This spread is hard to interpret because published evaluations score an end-to-end pipeline against a student baseline, which conflates three questions that can fail independently. Does the injection pipeline consume a foreign cache correctly (*mechanics*)? Does the translator replace the student’s own cache at replacement level (*mapping*)? Does the teacher’s cache carry content the frozen student can exploit at all (*exploitability*)? A positive delta could come from any of the three, and a null result cannot be attributed without measurements that separate them.

We therefore evaluate cache translation as an audit with three gated hypotheses, one per question. **H1 (mechanics)**: injecting the student’s own cache through the full translation-and-injection pipeline reproduces the student’s own prefill. **H2 (mapping)**: some tested translator reaches replacement level, scored by the change in gold-letter probability over the student’s own prefill. **H3 (exploitability)**: tested teacher-derived cache content provides an incremental benefit when introduced into an otherwise native student cache.

Our contribution is the audit itself and what it returns, not a new translator. Three results follow, and each is bounded by the designs we tested.

- **Mechanics is not the failure mode.** The identity control reproduces the student to numerical tolerance, and an offline/online pair that reloads only the persisted cache and translator agrees to the last recorded digit on the translated and identity paths.
- **No tested translator replaces the strong student’s cache.** Linear, per-layer, task-aware, architecture-anchored, and gradient-trained translators, calibration budgets from 10 to 500 examples, and the closest published design in both its averaged and concatenated form all fail: the largest point estimate on the strong pair is  $-0.138 [-0.312, +0.037]$  and every other interval lies below zero. The weak 0.6B student reaches  $+0.010 [-0.089, +0.106]$ , which establishes neither degradation nor our reporting margin.
- **No teacher-content probe establishes gain.** Mixing teacher-derived content into the student’s own cache degrades monotonically with the teacher fraction, replacing the top third of layers is neutral rather than helpful, and no probe configuration produces an interval that excludes zero on the positive side.

Two further results sharpen the negative. First, improving KV reconstruction does not improve answers: a joint MLP that reaches held-out  $R^2$  of 0.854 (keys) and 0.481 (values), against 0.810 and 0.323 for the per-head affine translator, still lands in the same CHG band. Second, repairing perplexity does not repair the answer deficit: a translator whose suffix perplexity is within 2.5 of the identity control answers 26.7% against the student’s 50.0%. Reconstruction fidelity and fluency are therefore not substitutes for task quality in this setting.

**Scope.** We measure a negative result about the designs we tested, not an impossibility claim. Our primary pair is one model family, four-choice classification with  $n$  between 10 and 100 evaluation samples per run, and the reporting margin  $\epsilon$  is not pre-registered (Section 3). Section 8 lists what this leaves open.

## 2 RELATED WORK

**KV reuse within one model.** Prompt caching, cache-augmented generation, and streaming attention reuse a model’s own cache: streaming attention identifies early positions with outsized attention mass (Xiao et al., 2023); CacheGen compresses and streams a cache (Liu et al., 2024); CacheBlend fuses cached fragments for retrieval-augmented generation (Yao et al., 2025); KVSharer shares caches across similar layers (Yang et al., 2024). These systems assume one model on both sides, so their question is storage and attention cost rather than representation compatibility.

**Cross-model cache translation.** C2C translates caches directly with a depth-ratio layer map (Fu et al., 2025); LSC aligns caches through a shared latent space (Dery et al., 2026); MoT combines gated translators over depth-ratio channel windows with a correction pass over the context (Lee et al., 2026); KVComm shares attention-ranked layers (Shi et al., 2025); Interlat communicates last hidden states (Du et al., 2025); HCache restores caches from hidden-state translations (Gao et al., 2025). The closest strict-zero-re-prefill design fits a closed-form linear map from selected source layers, with de-RoPE keys, and reports 73–98% retention on matched-KV pairs (Heo et al., 2026).

Two recent entries widen the range of outcomes: CacheBridge replaces full-head mapping with architecture-indexed support—each target head matched to one source head, with attention-aligned calibration—and reports 99.83% mean target retention on Qwen3 with an order of magnitude less calibration data (Qu et al., 2026); a universal context-reuse layer reports that translated KV states raise LongBench2 accuracy from 27.59% to 34.48% for Qwen2.5-7B→1.5B, 6.89 points above the receiver’s own prefill (Li et al., 2026). Retention measures how closely a translated cache reproduces the receiver’s own prefill, not whether the receiver gains over it, and neither of these reports isolates mechanics with an identity control. With the deficits we measure, the published evidence now contains near-native retention, above-baseline gain, and severe degradation at the same time. That spread is the case for a shared audit rather than for or against any one translator.

**Cross-model representation alignment.** Model stitching connects frozen networks with a learned layer (Bansal et al., 2021); relative representations enable zero-shot latent communication (Moschella et al., 2022); the Platonic representation hypothesis and its refinements argue for convergent representations across models (Huh et al., 2024; Ziyin & Chuang, 2025; Gröger et al., 2026); embedding-space methods align vocabularies post hoc (Jha et al., 2025). This line supports a shared geometry but does not measure whether it carries a specific task’s answer information.

**Frozen-model controllability.** Task vectors, activation steering, and tuned lenses show that a frozen model’s residual stream can be steered or read (Hendel et al., 2023; Stolfo et al., 2024; Belrose et al., 2023; Mu et al., 2023). Our probes are the KV-cache analogue of these interventions, and they come back empty on the pairs we test. FFN layers are key–value memories (Geva et al., 2021), which is one reason a weight-mediated capability gap need not survive as a cache-mediated one.

**What this line does not establish.** Across these lines, no cited evaluation (i) separates mechanics, mapping, and exploitability, (ii) injects the student’s own cache as an identity control, (iii) reports accuracy at matched perplexity, or (iv) constrains exploitability with a no-additional-training probe of teacher-derived content. This audit supplies those four measurements.

### 3 FRAMEWORK

#### 3.1 SETTING AND METRIC

A teacher  $T$  prefills a context  $x$  and stores a cache  $C_T(x) = \{K_\ell^T, V_\ell^T\}_{\ell=1}^{L_T}$  with  $H$  KV heads and head dimension  $d$ . A translator  $g_\theta$  maps it to  $\hat{C}_S(x)$  in the space of a student  $S$ . The student answers a query  $q$  from  $\hat{C}_S(x)$  alone, under strict zero re-prefill: no context token enters the student’s forward pass. Three references anchor every comparison: the *student self-prefill*  $S(q \mid C_S(x))$ , the *identity injection* that feeds  $C_S(x)$  through the injection pipeline, and the *teacher full-prefill* reference  $T(q \mid C_T(x))$ .

We score answer quality by the gold-letter probability  $p_{\text{gold}}$ : the softmax over the four answer-letter logits at the final prompt position, using each model’s own tokenizer id for each letter. Only four-choice examples are retained, so a student with no signal sits at 0.25. Throughout,  $CHG$  is the change in  $p_{\text{gold}}$  over the student’s own prefill on the same examples, with 95% bootstrap intervals. All intervals use a paired percentile bootstrap over evaluation examples with  $10^4$  resamples, including the family-wise probe bound in Section 5.3. Figure 1 summarizes the setting and the three gates that test it.

#### 3.2 THREE HYPOTHESES, THREE GATES

**H1 (mechanics).** The identity injection matches the student’s own prefill per sample. *Gate:* every evaluated sample has logit cosine  $\geq 0.9998$  against the student’s own prefill. The tolerance is orders of magnitude away from every non-identity method we observe, whose cosines sit near 0.5; we report the observed minimum alongside the gate so that the criterion is reproducible rather than an equality to one. H1 failing would mean the pipeline, not the translator, is broken.

**H2 (mapping).** Some tested translator reaches replacement level. *Gate:* the interval lower bound of  $CHG$  is above  $-\epsilon$ , with  $\epsilon = 0.02$  gold probability—a *reporting margin* selected after the gate form was frozen, about 4% of the 0.503 student baseline, and not a pre-registered constant. A stricter gain gate additionally requires the lower bound to exceed 0. We report two judgements per translated row,

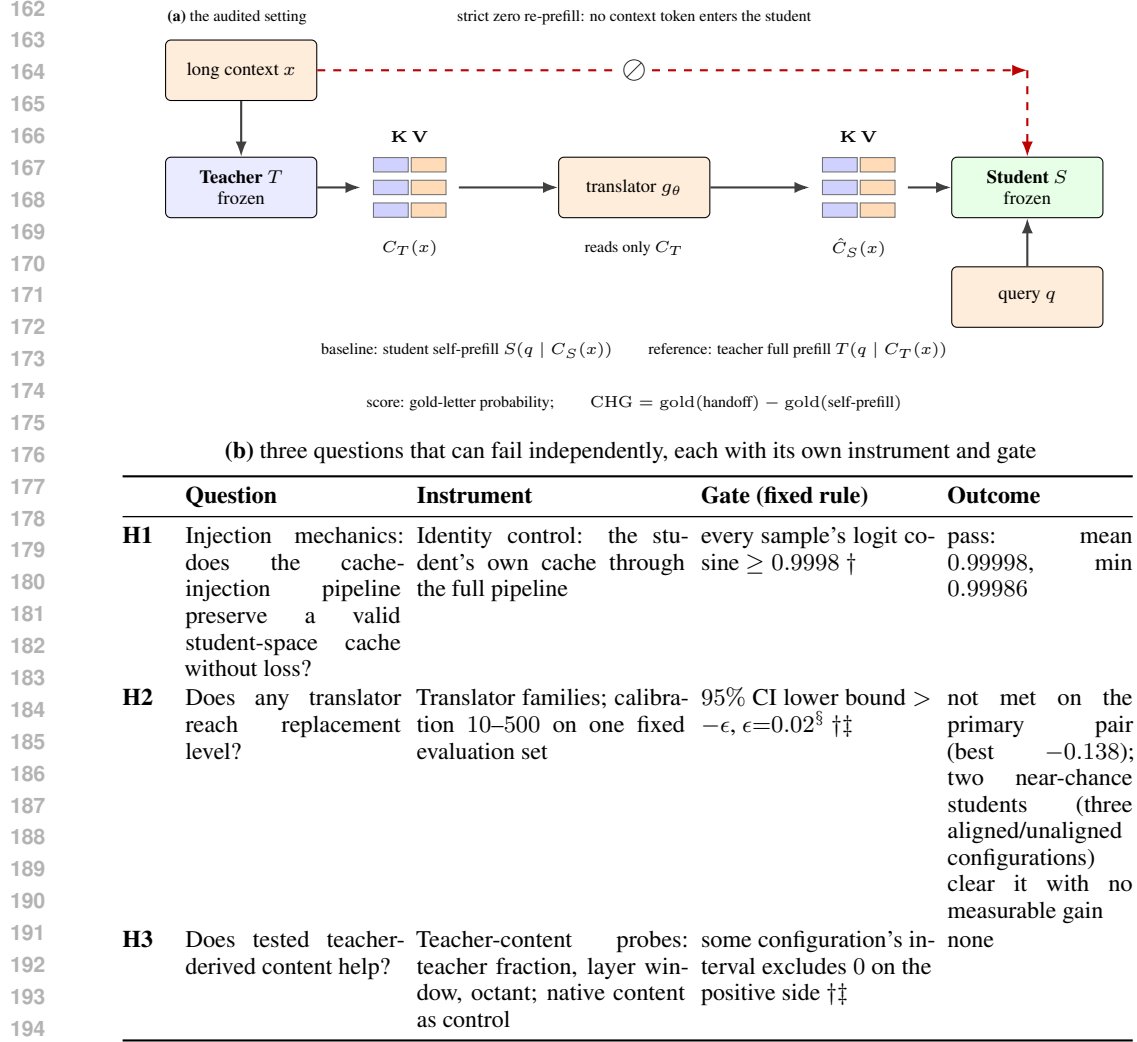


Figure 1: The audited setting and the three gated questions. (a) A teacher prefills a long context and stores its cache; a translator maps that cache into the student’s cache space; the student answers a short query from the translated cache alone, under strict zero re-prefill (red crossed path). The audit scores the handoff by the change in gold-letter probability against the student’s own prefill, with the teacher’s full prefill as a reference point. (b) The three questions can fail independently, so each has its own gate.

because they are conceptually independent: *replacement* (pass iff the interval lower bound is above  $-\epsilon$ , otherwise fail) and *direction* (degraded if the interval upper bound is below zero, gain if the lower bound is above zero, and neutral otherwise). A row can therefore be practically non-inferior and statistically worse than the baseline at the same time; Table 1 reports both judgements, and no row we record falls in that overlap.

**H3 (exploitability).** Tested teacher-derived cache content provides an incremental benefit when introduced into an otherwise native student cache. *Gate*: some probe configuration improves over the student’s own cache with an interval that excludes zero on the positive side. The probe does not train a translator, so a positive effect would show that teacher-derived content is exploitable under controlled interventions; a null result constrains the extraction paths we test without excluding every nonlinear one. The gate is therefore about tested content, not about the intrinsic information content of the teacher’s cache, and it is not specific to the end-to-end score of any single translator.

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### 3.3 EVIDENCE TIERS

The audit grew while revising, so we label every claim by how it entered the protocol. *Confirmatory* claims were fixed before the runs: the identity gate *in form* (its numerical tolerance 0.9998 is a revision-added operationalization, Appendix B), the non-inferiority *form* of the replacement gate and the  $\epsilon$ -free gain gate on the primary pair, and the fraction and window probes under the rule above. *Frozen-gate extensions* were added while revising and scored under the same gates and metrics: the residual-anchored and MLP translators, the 8B teacher, the calibration ladder, the cross-architecture and token-aligned students, the 1K retrieval task, the octant sweep, and the refit-variance study. *Exploratory* analyses were derived from the recorded runs afterwards and are worded as interpretations. We also report a joint bootstrap over the confirmatory family in Section 5.3, since no wider correction was pre-specified.

## 4 METHOD

**Models and pairs.** The primary pair is Qwen3-4B (teacher) to Qwen3-1.7B (student) (Qwen Team, 2025). We additionally run Qwen3-4B→Qwen3-0.6B, Qwen3-8B→Qwen3-1.7B, and Qwen3-8B→Qwen3-0.6B, plus five alternative students under the same protocol: Llama-3.2-1B, Llama-3.2-3B, Gemma-2-2B, Gemma-3-1B, and Qwen2.5-1.5B.

**Data.** Calibration and evaluation use four-choice items from HellaSwag and ARC with disjoint splits. The context-length check uses a four-choice needle-in-a-haystack task packed to a target length; the 4K and 8K runs store caches in float16, which we flag wherever those numbers appear.

**Protocol.** Every translated row is produced under zero re-prefill, with the identity control present in the same run, a unified prompt format across models, native rescaling of the injected cache, and per-run recording of the git hash, protocol version, and per-sample scores. The main evaluation set for the comparisons above is a fixed tail of 100 HellaSwag/ARC items; the translator ladder uses  $n = 30$  or  $n = 63$  items depending on the calibration budget, and we state  $n$  for every number we quote.

**Translators.** We test per-head ridge; per-head affine (centered, the strongest linear family); per-layer affine; a task-aware objective; a residual-anchored translator (RAT) built from the two models’ own projections and a shared vocabulary alignment, with component ablations; a per-head MLP; and a position-wise joint MLP of about 37M parameters. For the closest published design we implement top- $k$  source-layer selection with de-RoPE keys (Heo et al., 2026) in two forms: averaging the selected layers with 200 audit contexts, and concatenating them with 100 calibration sequences of 1,024 tokens from a generated web-text-style corpus; the reference uses 500 sequences (Appendix A).

To separate budget effects from evaluation items, we vary the calibration budget of the strongest linear family over  $c \in \{10, 30, 60, 100, 200, 500\}$  examples on the same fixed evaluation set.

**Probes.** The fraction probe injects  $\alpha\hat{C}_S + (1 - \alpha)C_S$  for  $\alpha \in \{0.25, 0.5, 0.75\}$ ; the window probe replaces the bottom, middle, or top third of student layers with translated content; the octant probe replaces one of eight layer windows. All probes require the student’s own prefill, so they are measurement tools, not deployable systems.

**Diagnostics.** We measure held-out reconstruction  $R^2$  of the student’s own  $K$  and  $V$  from the teacher’s states with the same fit budget for every family (20 calibration contexts, 10 disjoint held-out contexts), and we report KV norm ratios between teacher and student.

## 5 RESULTS

### 5.1 H1: MECHANICS IS NOT THE FAILURE MODE

The identity injection reproduces the student’s own prefill: on the primary pair the change is  $+0.0001$   $[-0.0006, +0.0009]$  with per-sample logit cosine 0.99998 on average and 0.99986 at worst ( $n=30$ ), and the control holds in every other run we report, whose mean cosines lie between 0.99998 and 0.99999. The guarantee also survives persistence: an online run that reloads only the persisted cache and translator, never the teacher, reproduces the offline translated and identity results at the recorded

precision ( $-0.1376$  and  $+0.00013$ ). The native-cache ablation differs between the two runs because the online path carries no native cache, so we scope equivalence to the translated and identity paths. Any deficit below is therefore attributable to the translator or to the content it moves.

## 5.2 H2: NO TESTED TRANSLATOR REPLACES THE STRONG STUDENT’S CACHE

Table 1 collects the translator ladder. Every strong-student translation has a negative point estimate relative to the student’s own prefill, from  $-0.138$  to  $-0.392$ , and all but the best configuration are statistically degraded. The best configuration (per-head affine,  $c=30$ ) is the only one whose interval reaches zero ( $-0.138 [-0.312, +0.037]$ ), and it does not pass the replacement gate. Three controls show that this is not an artifact of our implementation choices.

*Calibration volume does not close the gap in the tested range.* On one fixed evaluation set the same family moves between  $-0.2569$  and  $-0.2378$  across the six budgets, a spread of  $0.02$  against intervals of width  $\geq 0.2$ .

*Capacity and anchoring do not close the gap in the tested range.* A per-head MLP (five trainings, range  $-0.39$  to  $-0.26$ ) and a 37M-parameter joint MLP (five trainings,  $-0.30$  to  $-0.26$ ) sit inside the linear band; removing RAT’s analytic anchor raises suffix perplexity from  $23.7$  to  $501.6$  while changing CHG from  $-0.223$  to  $-0.268$ —fluency moves, the answer deficit does not.

*Source-layer combination does not close the gap in the tested range.* The closest published design fails in both forms we can match—its source-layer combination and its sequence length—while differing in calibration volume and corpus (Appendix A): averaging selected top- $k$  layers gives  $-0.273$ ,  $-0.308$ , and  $-0.291$  for  $k=1, 3, 5$ ; concatenating them with long-sequence web-text calibration gives  $-0.279$ ,  $-0.223$ , and  $-0.254$ . All six intervals lie below zero, and the identity control passes in all six runs.

*The weak student is a different story, but not a positive one.* It reaches  $+0.010 [-0.089, +0.106]$ , whose interval spans both zero and  $-\epsilon$ : we establish neither replacement nor degradation. Two further weak-student rows behave the same way ( $-0.023 [-0.106, +0.060]$  with an 8B teacher;  $-0.041 [-0.124, +0.038]$  for RAT). The largest point estimate anywhere in the audit is  $+0.010$ , and it belongs to this pair.

## 5.3 H3: NO TEACHER-CONTENT PROBE ESTABLISHES GAIN

Appendix Table 5 reports the probes under the confirmatory translator (RAT,  $c=30$ ) together with a replication under the strongest linear translator (affine,  $c=30$ ).

The fraction probe degrades monotonically with the teacher fraction under both translators, so adding teacher-derived content to the student’s own cache does not help. The window probe shows a real asymmetry, but not a useful one: replacing the top third of layers is neutral (its interval contains zero in both translators), while the middle and bottom thirds are harmful; the ordering between middle and bottom is translator-dependent, so we claim only that the top window is neutral.

Over the six confirmatory configurations (three fractions, three windows, RAT source) we bootstrap the maximum of the mean CHG over evaluation samples ( $10^4$  resamples). The observed maximum is  $-0.001$  and the 95% upper estimate for the maximum gain is  $+0.009$ . This is a percentile bound on a maximum, not a centered max- $t$  construction; we therefore read it as “the bootstrap 95% upper estimate of the best achievable gain in this family is about one gold-probability point”, and not as a simultaneous 95% exclusion statement. Under the affine replication every configuration is again non-positive, and a best-of-eight octant sweep reaches  $+0.020 [-0.012, +0.052]$ , whose interval contains zero. Across all 48 probe rows recorded under the protocol, none has an interval that lies entirely above zero.

## 6 DECOUPLINGS AND THE REACH OF THE DEFICIT

Two of the audit’s findings are inequalities rather than mechanisms, and both come directly from the recorded runs: better reconstruction does not buy better answers, and neither does better perplexity.

Table 1: Translator ladder on the primary pair unless noted. Acc/Gold/PPL: answer accuracy, gold-letter probability, and suffix perplexity of the translated cache; the student self-prefill row’s perplexity is measured over the full context and is not directly comparable. CHG is the gold-probability change over the student’s own prefill, with 95% bootstrap intervals; ranges are over independent trainings. *Repl.* is the replacement judgement of Section 3 (pass iff the interval lower bound exceeds  $-\epsilon$ ,  $\epsilon=0.02$ ). *Dir.* is the direction (*degraded* iff the interval upper bound is below zero, *gain* iff the lower bound exceeds zero, *neutral* otherwise). The two judgements are independent, so a row can be practically non-inferior while being significantly worse than the baseline. The identity row is scored by the H1 gate.  $n$  is the evaluation sample count. Rows marked  $\ddagger$  are frozen-gate extensions;  $\#$  concatenates selected source layers and calibrates on 1,024-token web-text passages.

| Configuration                                          | Acc / Gold / PPL                    | Gold CHG [95% CI]          | $n$ | Repl. | Dir.     |
|--------------------------------------------------------|-------------------------------------|----------------------------|-----|-------|----------|
| Identity control (self-KV), 1.7B                       | 0.500 / 0.503 / 21.2                | +0.0001 [−0.0006, +0.0009] | 30  | pass  | neutral  |
| Student self-prefill, 1.7B                             | 0.500 / 0.503 / 25.4                | —                          | 30  | —     | —        |
| Teacher full prefill, 1.7B                             | 0.767 / 0.693 / —                   | —                          | 30  | —     | —        |
| Native cache injection, 1.7B                           | 0.167 / 0.169 / $5.4 \times 10^5$   | −0.334 [−0.510, −0.149]    | 30  | fail  | degraded |
| Ridge per-head, c30, 1.7B                              | 0.200 / 0.207 / 88.5                | −0.296 [−0.470, −0.112]    | 30  | fail  | degraded |
| Affine per-head, c30, 1.7B                             | 0.367 / 0.365 / 46.1                | −0.138 [−0.312, +0.037]    | 30  | fail  | neutral  |
| Affine per-head, c200, 1.7B                            | 0.254 / 0.258 / 56.4                | −0.239 [−0.353, −0.122]    | 63  | fail  | degraded |
| Affine per-layer, c200, 1.7B                           | 0.206 / 0.222 / $8.1 \times 10^4$   | −0.275 [−0.404, −0.140]    | 63  | fail  | degraded |
| Task-aware, c200, 1.7B                                 | 0.270 / 0.280 / 142.1               | −0.216 [−0.336, −0.093]    | 63  | fail  | degraded |
| RAT, c30, 1.7B $\ddagger$                              | 0.233 / 0.235 / 501.6               | −0.268 [−0.439, −0.092]    | 30  | fail  | degraded |
| RAT, c200, 1.7B $\ddagger$                             | 0.238 / 0.276 / 1008.2              | −0.221 [−0.342, −0.099]    | 63  | fail  | degraded |
| RAT no-anchor rank-32, c30 $\ddagger$                  | 0.267 / 0.280 / 23.7                | −0.223 [−0.369, −0.086]    | 30  | fail  | degraded |
| Per-head MLP, c30 (5 trainings) $\ddagger$             | 0.133–0.267 / 0.111–0.246 / 65–96   | −0.39 to −0.26             | 30  | fail  | degraded |
| Per-head MLP, c200 (5 trainings) $\ddagger$            | 0.175–0.302 / 0.241–0.271 / 109–149 | −0.26 to −0.23             | 63  | fail  | degraded |
| Joint MLP, c30 (5 trainings) $\ddagger$                | 0.167–0.233 / 0.201–0.239 / 26–32   | −0.30 to −0.26             | 30  | fail  | degraded |
| Top- $k$ ridge (avg), c200, $k=1$ $\ddagger$           | 0.240 / 0.238 / 78.5                | −0.273 [−0.386, −0.159]    | 100 | fail  | degraded |
| Top- $k$ ridge (avg), c200, $k=3$ $\ddagger$           | 0.190 / 0.203 / 2630                | −0.308 [−0.414, −0.199]    | 100 | fail  | degraded |
| Top- $k$ ridge (avg), c200, $k=5$ $\ddagger$           | 0.210 / 0.220 / $6.9 \times 10^4$   | −0.291 [−0.396, −0.184]    | 100 | fail  | degraded |
| Top- $k$ ridge (concat, web-text), $k=1$ $\ddagger$    | 0.230 / 0.233 / $1.6 \times 10^5$   | −0.279 [−0.407, −0.148]    | 100 | fail  | degraded |
| Top- $k$ ridge (concat, web-text), $k=3$ $\ddagger$    | 0.290 / 0.289 / $4.6 \times 10^4$   | −0.223 [−0.364, −0.081]    | 100 | fail  | degraded |
| Top- $k$ ridge (concat, web-text), $k=5$ $\ddagger$    | 0.270 / 0.257 / 3678                | −0.254 [−0.391, −0.117]    | 100 | fail  | degraded |
| Ridge per-head, c500, 8B $\rightarrow$ 1.7B $\ddagger$ | 0.310 / 0.288 / 146.1               | −0.224 [−0.327, −0.122]    | 100 | fail  | degraded |
| Affine per-head, c200, 4B $\rightarrow$ 0.6B           | 0.302 / 0.283 / 71.8                | +0.010 [−0.089, +0.106]    | 63  | fail  | neutral  |
| Affine per-head, c30, 8B $\rightarrow$ 0.6B $\ddagger$ | 0.270 / 0.269 / 99.0                | −0.023 [−0.106, +0.060]    | 100 | fail  | neutral  |
| RAT, c200, 4B $\rightarrow$ 0.6B $\ddagger$            | 0.206 / 0.232 / 790.0               | −0.041 [−0.124, +0.038]    | 63  | fail  | neutral  |
| Affine per-head, c30, 1K retrieval $\ddagger$          | 0.233 / 0.232 / 6554                | −0.285 [−0.549, −0.003]    | 30  | fail  | degraded |
| Affine per-head, c10, 4K retrieval $\ddagger$          | 0.200 / 0.200 / $1.6 \times 10^4$   | −0.596 [−0.939, −0.135]    | 10  | fail  | degraded |
| Affine per-head, c10, 8K retrieval $\ddagger$          | 0.200 / 0.203 / $2.4 \times 10^4$   | −0.577 [−0.943, −0.082]    | 10  | fail  | degraded |

## 6.1 RECONSTRUCTION FIDELITY DOES NOT IMPLY FUNCTIONAL EQUIVALENCE

Appendix Table 4 reports held-out reconstruction under one fixed budget (20 calibration contexts, 10 disjoint held-out contexts) for four configurations.

Two contrasts carry the point. The joint MLP reaches  $R_V^2 = 0.481$  against the per-head affine translator’s 0.323 and still produces a CHG between −0.30 and −0.26, inside the band of every other strong-student translator. The choice of source-layer combination moves reconstruction in opposite directions (averaging degrades it with  $k$ ; concatenation improves it) while leaving the verdict unchanged. Whatever the translators recover does not appear to include the part of the cache the student needs in order to answer.

## 6.2 FLUENCY DOES NOT IMPLY CAPABILITY

The same pattern appears on perplexity. Our most fluent failing translator reaches suffix perplexity 23.7, within 2.5 of the identity control’s 21.2, and still answers 26.7% against the student’s 50.0%; the joint MLP is fluent too (31.6) with gold probability 0.238 against 0.503. Perplexity tracks whether the cache permits a fluent continuation, accuracy tracks whether it supports the task, and the two dissociate across the recorded runs—which matters because perplexity is cheap and commonly reported as a quality proxy.

### 6.3 CHANNEL ASYMMETRY

At  $c=30$  with the ridge family, mapping only values answers 36.7% of items, mapping only keys answers 30.0%, and mapping both answers 20.0%, while suffix perplexity orders the three in the opposite direction ( $1.85 \times 10^7$ , 253, and 88.5). Mapping keys therefore repairs fluency while damaging task behavior. We report this as an observation about the tested channel translators. Without attention-map analysis we do not claim to have identified the mechanism; the reading most consistent with the pattern is that key-side routing errors produce the asymmetry.

### 6.4 OTHER CONTEXTS, STUDENTS, AND REPAIR ATTEMPTS

The deficit is not specific to short contexts. A four-choice retrieval task packed to roughly 1K tokens gives  $-0.285 [-0.549, -0.003]$  ( $n=30$ ); with caches stored in float16 and the fixed-tail protocol disabled, a true 4K run gives  $-0.596 [-0.939, -0.135]$  and a true 8K run gives  $-0.577 [-0.943, -0.082]$  (both  $n=10$ ). These two rows are a different evaluation regime from the 1K row, and we read them as further degradation rather than as a calibrated length sweep.

The deficit is also not specific to one architecture family: five alternative students give three negative rows (Llama-3.2-3B  $-0.044$ , Gemma-3-1B  $-0.061$ , Qwen2.5-1.5B  $-0.157$ ; all  $n=100$ ) and two near-parity rows on students close to chance (Llama-3.2-1B  $+0.000$ , Gemma-2-2B  $-0.009$ ), while token alignment leaves the ranking unchanged (Appendix C). A naive target-side re-read, which violates zero re-prefill, does not recover the student either ( $-0.232 [-0.335, -0.134]$  against  $-0.249 [-0.353, -0.149]$ ); we did not re-implement any published correction loss, so this bounds a naive replay rather than those systems.

A text-channel control points the same way: summarizing the teacher’s context into text reaches 0.440 accuracy and 0.404 gold against the student’s 0.520 and 0.512, a difference of  $-0.108$  ( $n=100$ ).

Over all rows recorded under the protocol, the largest translation point estimate is  $+0.010 [-0.089, +0.106]$ , from the weak-student pair, and no probe row places its interval above zero (Appendix B).

## 7 IMPLICATIONS FOR CACHE-TRANSLATION RESEARCH

Three checks would have caught what an end-to-end score hides. Report an identity control that injects the student’s own cache through the full pipeline, because it separates a broken pipeline from a broken translator. Report accuracy, not only retention or perplexity, because retention measures similarity to the receiver’s own prefill rather than the receiver’s gain over it, and because fluency and task quality dissociate in our runs. Report a no-additional-training probe of teacher-derived content before investing in translator capacity: the probe takes its content from a fixed, already-calibrated translator and trains nothing new, and in our audit no probe configuration produces an interval that excludes zero.

The decouplings above carry the scientific weight: reconstruction quality and perplexity are not substitutes for task quality, so a translation result reported only through them is underdetermined. The reading that the teacher’s advantage is mediated by parameter-dependent computation is consistent with our results, since FFN layers act as key-value memories, but we present it as an interpretation rather than an impossibility result: a translator outside our tested families might behave differently.

## 8 LIMITATIONS

Our scale pairs all come from one model family on the primary axis, with  $n$  between 10 and 100 evaluation samples per run, so the intervals are wide. The 4K and 8K retrieval rows use  $n=10$ , a float16 cache regime, and a different evaluation protocol, so they qualify the context-length claim rather than establishing a length sweep.

**One primary pair.** The ladder, the calibration study, and the probes are all run on Qwen3-4B $\rightarrow$ 1.7B. The cross-architecture check covers five alternative students but three of them are near chance, so it establishes mechanics and gate-level replacement rather than capability.

**Exploitability is tested, not exhausted.** Our probes cover linear blending, layer-window replacement, and an octant sweep under two translators. A nonlinear or jointly structured extraction path outside this family would not be detected, and the null result is evidence about the tested interventions rather than a proof of impossibility. A positive probe result would have been decisive; ours is not.

**Gradient-trained families are seed-dependent and not always reproducible.** The per-head MLP is the clearest case: two runs with the same configuration and the same recorded seed differ by 0.046 in CHG ( $-0.303$  versus  $-0.257$ ). We therefore report those families as ranges over trainings, and we do not treat any single MLP number as a reproducible point estimate. The joint MLP reproduces its seed to within 0.001.

**The reporting margin is not pre-registered.**  $\epsilon=0.02$  was chosen after the gate form was frozen, as a practical reporting margin; the sensitivity of our conclusions to that choice is reported in Appendix B.

Some measurements remain thin. The cold-start equivalence rests on a single offline/online pair and covers the translated and identity paths only. The held-out reconstruction  $R^2$  comes from one fit of 20 contexts with 10 held-out contexts, so it constrains rather than characterizes what a translator can recover. Our implementation of the closest published design matches the reference’s combination and sequence length; it uses 100 calibration sequences instead of 500, drawn from a generated corpus. We report no deployment-cost claim: the recorded pipeline measure includes a measurement-only forward pass and is negative ( $-9.7$  at best).

## 9 CONCLUSION

Under strict zero re-prefill, no tested translator establishes replacement for our primary strong-student pair, and the only rows that clear the reporting replacement margin sit on near-chance alternative students and show no measurable gain. The injection pipeline itself is exact to numerical tolerance, so the failure is not mechanical: the deficit on the primary pair persists across the tested calibration budgets, translator capacities, source-layer combinations, and token-alignment controls; and it is not repaired by matching the closest published design’s source-layer composition and calibration sequence length under our generated calibration corpus and 100-sequence volume. The two findings we expect to outlast this setting are the decouplings: cache reconstruction quality and cache-induced fluency are not substitutes for task quality. We release the runs, the protocol, and the three checks so that future translation claims can be checked against all of them.

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Table 2: Our implementation of the closest strict-zero-re-prefill design (Heo et al., 2026) and where it differs from the reference. Both configurations of our implementation fail (Table 1).

| Aspect                   | This audit                                              | Reference            |
|--------------------------|---------------------------------------------------------|----------------------|
| Source-layer combination | concatenation (aligned); averaged variant also reported | concatenation        |
| Key mapping              | de-RoPE keys                                            | yes (de-RoPE keys)   |
| Penalty                  | $\lambda = 10^{-3}$ , $K$ and $V$ fitted separately     | not specified        |
| Candidate source layers  | all teacher layers                                      | not specified        |
| Calibration corpus       | generated web-text-style 1,024-token passages           | FineWeb-Edu web text |
| Calibration volume       | 100 sequences                                           | 500 sequences        |

Table 3: Gradient-trained families: interval detail for the rows Table 1 reports as training ranges ( $10^4$ -draw paired bootstrap). Every recorded training of every family has an interval upper bound below zero, which is what its *degraded* direction label requires.

| Family                | Trainings | Point estimates  | Largest CI upper | Per-training CI uppers                     |
|-----------------------|-----------|------------------|------------------|--------------------------------------------|
| Per-head MLP, $c=30$  | 5         | -0.392 to -0.257 | -0.069           | -0.148/ - 0.069/ - 0.136/ - 0.231/ - 0.167 |
| Per-head MLP, $c=200$ | 5         | -0.256 to -0.226 | -0.104           | -0.104/ - 0.139/ - 0.144/ - 0.104/ - 0.118 |
| Joint MLP, $c=30$     | 5         | -0.302 to -0.264 | -0.095           | -0.115/ - 0.095/ - 0.128/ - 0.120/ - 0.143 |

## A IMPLEMENTATION DIFFERENCES FROM THE CLOSEST PUBLISHED DESIGN

## B REPRODUCIBILITY AND RECORDED INVENTORY

Every number in this paper comes from a recorded run directory containing the run’s configuration, per-sample scores, the git hash, the protocol version, and a zero-re-prefill verification flag. At the time of writing the repository contains 177 evaluation runs with metrics and 181 configuration rows under the audit protocol (133 translations, 48 probes). A re-run audit of 23 core configurations reproduced the deterministic families to within  $|\Delta| \leq 0.002$  in CHG; the per-head MLP family did not reproduce across runs with the same seed (Section 8), which is why it is reported as a range. For the gradient-trained families the runs set Python, NumPy, and PyTorch seeds (CPU and CUDA) but do not enable deterministic kernel selection, so the residual source of nondeterminism has not been isolated.

The margin sensitivity referenced in Section 3 is recorded in the same directory: conclusions about the strong-student pair are unchanged for  $\epsilon \in \{0.01, 0.02, 0.05\}$  because their intervals lie below zero, while the weak-student pair fails the replacement gate under all three.

Intervals are recomputed from those per-sample scores with  $10^4$  paired percentile bootstrap resamples (scripts/bootstrap\_ci\_report.py, artifact reports/bootstrap\_1e4/ci\_1e4.json); an earlier draft’s  $10^3$ -draw endpoints differ by up to 0.02, the Monte-Carlo noise of  $10^3$  draws at these sample sizes, so the  $10^4$ -draw values are reported. The identity gate’s *form* was fixed before the runs, while the numerical tolerance 0.9998 was introduced during revision as a reproducible operationalization of the original “within numerical tolerance” criterion; the recorded runs store logit cosines rather than logits, so the residual identity difference (minimum 0.99986) is not decomposed further.

## C ADDITIONAL MEASUREMENTS

**Cross-architecture and token-aligned rows.** Under the rectangular translator (head mean-pooling plus dimension projection),  $n=100$ : Llama-3.2-1B +0.000 [-0.012, +0.011], Gemma-2-2B -0.009 [-0.020, +0.002], Llama-3.2-3B -0.044 [-0.074, -0.013], Gemma-3-1B -0.061 [-0.132, +0.006], Qwen2.5-1.5B -0.157 [-0.212, -0.106]. With character-overlap token alignment applied to the two tokenizers that differ most: Llama-3.2-3B -0.024 [-0.047, -0.002], Gemma-3-1B -0.073 [-0.131, -0.018], Llama-3.2-1B -0.003 [-0.012, +0.006], Gemma-2-2B -0.007 [-0.018, +0.004]. Alignment changes neither the replacement verdict nor the absence of capability gain; for the shared-tokenizer student it is a no-operation by construction.

**Octant sweep.** Under the affine translator at  $c=30$ , the best of eight layer windows reaches  $+0.020$   $[-0.012, +0.052]$  ( $n=30$ ); no window’s interval lies above zero.

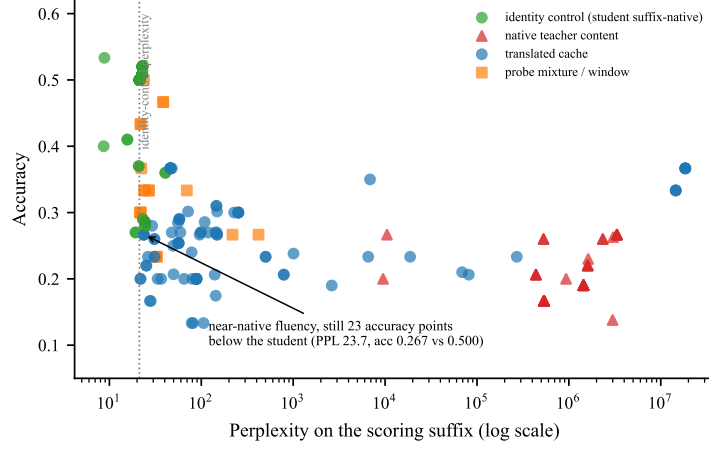


Figure 2: Perplexity over the scoring suffix against accuracy for every recorded method and run (log scale). The identity control—the student’s suffix-native reference—sits in the top-left; translated caches spread across a broad band, including a point with near-native perplexity (23.7) whose accuracy stays 23 points below the student’s 0.500. Perplexity does not order methods the way the task metric does.

Table 4: Held-out  $R^2$  for recovering the student’s own  $K$  and  $V$  from the teacher’s states (per-(layer, head) macro-average; training-split values in parentheses where recorded). Better reconstruction does not move the task verdict: the joint MLP nearly doubles the value-channel  $R^2$  of the affine translator and lands in the same CHG band (Table 1).

| Translator                  | Source-layer combination | $R_K^2$               | $R_V^2$               | Downstream CHG                 |
|-----------------------------|--------------------------|-----------------------|-----------------------|--------------------------------|
| Affine per-head, $c=30$     | –                        | 0.810                 | 0.323                 | $-0.138 [-0.312, +0.037]$      |
| Joint MLP, $c=30$           | all layers               | 0.854 (0.949)         | 0.481 (0.647)         | $-0.30$ to $-0.26$             |
| Top- $k$ ridge, $k=1, 3, 5$ | averaged                 | 0.822 / 0.787 / 0.762 | 0.317 / 0.248 / 0.182 | $-0.273$ / $-0.308$ / $-0.291$ |
| Top- $k$ ridge, $k=1, 3, 5$ | concatenated             | 0.826 / 0.858 / 0.864 | 0.326 / 0.419 / 0.428 | $-0.279$ / $-0.223$ / $-0.254$ |

Table 5: Teacher-content probes on the primary pair,  $n=30$ . The gold column is the gold-letter probability of the mixture; CHG is against the student’s own prefill. The first row is the student’s own cache. Both translators agree that every probe degrades or is neutral, and that replacing the top third of layers is neutral.

| Probe                             | Gold (RAT c30) | CHG (RAT c30)                | Gold (affine c30) | CHG (affine c30)             |
|-----------------------------------|----------------|------------------------------|-------------------|------------------------------|
| Student cache only ( $\alpha=0$ ) | 0.503          | $+0.0001 [-0.0006, +0.0009]$ | 0.503             | $+0.0001 [-0.0006, +0.0009]$ |
| Fraction $\alpha=0.25$            | 0.380          | $-0.123 [-0.258, +0.009]$    | 0.417             | $-0.086 [-0.183, +0.003]$    |
| Fraction $\alpha=0.50$            | 0.279          | $-0.224 [-0.374, -0.077]$    | 0.356             | $-0.147 [-0.322, +0.026]$    |
| Fraction $\alpha=0.75$            | 0.249          | $-0.254 [-0.423, -0.083]$    | 0.292             | $-0.211 [-0.400, -0.012]$    |
| Window: top third                 | 0.502          | $-0.001 [-0.010, +0.010]$    | 0.487             | $-0.016 [-0.120, +0.080]$    |
| Window: middle third              | 0.265          | $-0.238 [-0.383, -0.106]$    | 0.244             | $-0.259 [-0.413, -0.112]$    |
| Window: bottom third              | 0.224          | $-0.279 [-0.445, -0.105]$    | 0.321             | $-0.182 [-0.306, -0.070]$    |

**Family-wise detail.** The confirmatory family comprises three mixture fractions and three layer windows under the RAT translator. The joint bootstrap over  $10^4$  resamples gives an observed maximum of  $-0.001$  and a 95% upper estimate of  $+0.009$ .

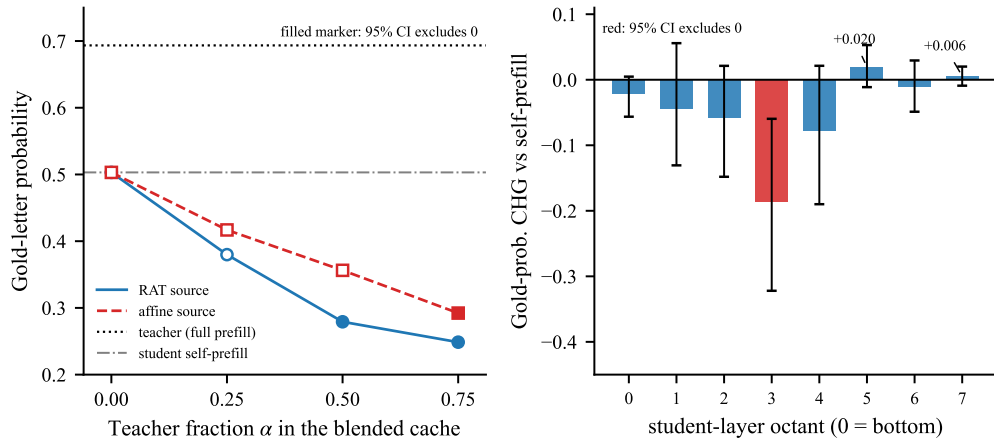


Figure 3: Teacher-content probes on the primary pair ( $n=30$ ). Left: gold-letter probability as translated content replaces the student’s own cache, for three mixture fractions under both translators. Right: layer windows. Every configuration degrades or is neutral relative to the student’s own prefill; the top-third window is the only neutral one. These are the rows of Table 5.